Olympiad Algebraic Manipulation, Vieta & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01 • Core International Olympiad Pillars

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Olympiad Algebraic Manipulation, Vieta & Inequalities Practice Exercises

1 Let x be a positive real number satisfying $x + \frac{1}{x} = 3$ (illustrated on the positive branch of $g(x) = x + 1/x$). Without solving for x, find the exact value of $x^3 + \frac{1}{x^3}$.

a. 27

b. 36

c. 18 ✓

d. 15

Let $S_n = x^n + x^{-n}$ with $S_1 = 3$. Squaring gives $S_2 = S_1^2 - 2 = 3^2 - 2 = 7$. Using $S_3 = S_1 S_2 - S_1 = S_1^3 - 3S_1$, we obtain $3^3 - 3(3) = 18$.

2 Let x be a positive real number satisfying $x + \frac{1}{x} = 4$ (illustrated on the positive branch of $g(x) = x + 1/x$). Without solving for x, find the exact value of $x^4 + \frac{1}{x^4}$.

a. 196

b. 194 ✓

c. 198

d. 190

Let $S_n = x^n + x^{-n}$ with $S_1 = 4$. Squaring gives $S_2 = S_1^2 - 2 = 4^2 - 2 = 14$. Squaring S_2 gives $S_4 = S_2^2 - 2 = 14^2 - 2 = 194$.

3 Let x be a positive real number satisfying $x + \frac{1}{x} = 5$ (illustrated on the positive branch of $g(x) = x + 1/x$). Without solving for x, find the exact value of $x^3 + \frac{1}{x^3}$.

a. 110 ✓

b. 125

c. 140

d. 105

Let $S_n = x^n + x^{-n}$ with $S_1 = 5$. Squaring gives $S_2 = S_1^2 - 2 = 5^2 - 2 = 23$. Using $S_3 = S_1 S_2 - S_1 = S_1^3 - 3S_1$, we obtain $5^3 - 3(5) = 110$.

© ClipSAT — clipsat.org | All Rights Reserved

4 Let x be a positive real number satisfying $x + \frac{1}{x} = 6$ (illustrated on the positive branch of $g(x) = x + 1/x$). Without solving for x, find the exact value of $x^4 + \frac{1}{x^4}$.

a. 1156

b. 1158

c. 1148

d. 1154 ✓

Let $S_n = x^n + x^{-n}$ with $S_1 = 6$. Squaring gives $S_2 = S_1^2 - 2 = 6^2 - 2 = 34$. Squaring S_2 gives $S_4 = S_2^2 - 2 = 34^2 - 2 = 1154$.

5 Let r, s, t be the three complex roots of the cubic polynomial $P(z) = z^3 - 3z^2 + 2z - 1$. Using Newton's power-sum recurrences, find $r^3 + s^3 + t^3$.

a. 9

b. 15

c. 12 ✓

d. 13

By Vieta's relations, $e_1 = 3,\ e_2 = 2,\ e_3 = 1$. First $p_2 = e_1^2 - 2e_2 = 3^2 - 2(2) = 5$. By Newton's third identity $p_3 = e_1^3 - 3e_1e_2 + 3e_3 = 3^3 - 3(3)(2) + 3(1) = 12$.

6 Let r, s, t be the three complex roots of the cubic polynomial $P(z) = z^3 - 4z^2 + 3z - 2$. Using Newton's power-sum recurrences, find $r^3 + s^3 + t^3$.

a. 28

b. 34 ✓

c. 40

d. 35

By Vieta's relations, $e_1 = 4,\ e_2 = 3,\ e_3 = 2$. First $p_2 = e_1^2 - 2e_2 = 4^2 - 2(3) = 10$. By Newton's third identity $p_3 = e_1^3 - 3e_1e_2 + 3e_3 = 4^3 - 3(4)(3) + 3(2) = 34$.

7 Let r, s, t be the three complex roots of the cubic polynomial $P(z) = z^3 - 5z^2 + 4z - 3$. Using Newton's power-sum recurrences, find $r^3 + s^3 + t^3$.

a. 74 ✓

b. 65

c. 85

d. 75

By Vieta's relations, $e_1 = 5,\ e_2 = 4,\ e_3 = 3$. First $p_2 = e_1^2 - 2e_2 = 5^2 - 2(4) = 17$. By Newton's third identity $p_3 = e_1^3 - 3e_1e_2 + 3e_3 = 5^3 - 3(5)(4) + 3(3) = 74$.

© ClipSAT — clipsat.org | All Rights Reserved

8 Let r, s, t be the three complex roots of the cubic polynomial $P(z) = z^3 - 6z^2 + 5z - 4$. Using Newton's power-sum recurrences, find $r^3 + s^3 + t^3$.

a. 126

b. 156

c. 139

d. 138 ✓

By Vieta's relations, $e_1 = 6,\ e_2 = 5,\ e_3 = 4$. First $p_2 = e_1^2 - 2e_2 = 6^2 - 2(5) = 26$. By Newton's third identity $p_3 = e_1^3 - 3e_1e_2 + 3e_3 = 6^3 - 3(6)(5) + 3(4) = 138$.

9 Let x, y, z be positive real numbers satisfying $x + y + z = 1$. Find the exact minimum value of $\frac{1}{x} + \frac{4}{y} + \frac{9}{z}$.

a. 14

b. 42

c. 36 ✓

d. 34

By Titu's Lemma (Engel's form of Cauchy-Schwarz), for $x, y, z > 0$: $\frac{1^2}{x} + \frac{2^2}{y} + \frac{3^2}{z} \geq \frac{(1+2+3)^2}{x+y+z} = \frac{6^2}{1} = 36$. Equality is achieved when $\frac{x}{1} = \frac{y}{2} = \frac{z}{3} = \frac{1}{6}$.

10 Let x, y, z be positive real numbers satisfying $x + y + z = 7$. Find the exact minimum value of $\frac{1}{x} + \frac{4}{y} + \frac{16}{z}$.

a. 21

b. 7 ✓

c. 14

d. 5

By Titu's Lemma (Engel's form of Cauchy-Schwarz), for $x, y, z > 0$: $\frac{1^2}{x} + \frac{2^2}{y} + \frac{4^2}{z} \geq \frac{(1+2+4)^2}{x+y+z} = \frac{7^2}{7} = 7$. Equality is achieved when $\frac{x}{1} = \frac{y}{2} = \frac{z}{4} = \frac{7}{7}$.

11 Let x, y, z be positive real numbers satisfying $x + y + z = 1$. Find the exact minimum value of $\frac{1}{x} + \frac{9}{y} + \frac{16}{z}$.

a. 64 ✓

b. 26

c. 72

d. 61

By Titu's Lemma (Engel's form of Cauchy-Schwarz), for $x, y, z > 0$: $\frac{1^2}{x} + \frac{3^2}{y} + \frac{4^2}{z} \geq \frac{(1+3+4)^2}{x+y+z} = \frac{8^2}{1} = 64$. Equality is achieved when $\frac{x}{1} = \frac{y}{3} = \frac{z}{4} = \frac{1}{8}$.

© ClipSAT — clipsat.org | All Rights Reserved

12 Let x, y, z be positive real numbers satisfying $x + y + z = 9$. Find the exact minimum value of $\frac{4}{x} + \frac{9}{y} + \frac{16}{z}$.

a. 29

b. 18

c. 6

d. 9 ✓

By Titu's Lemma (Engel's form of Cauchy-Schwarz), for $x, y, z > 0$: $\frac{2^2}{x} + \frac{3^2}{y} + \frac{4^2}{z} \geq \frac{(2+3+4)^2}{x+y+z} = \frac{9^2}{9} = 9$. Equality is achieved when $\frac{x}{2} = \frac{y}{3} = \frac{z}{4} = \frac{9}{9}$.

© ClipSAT — clipsat.org | All Rights Reserved